

Data Analysis

Automobile Dataset Analysis

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1 Dataset description

1.1 Structure

The dataset under study contains information about passenger vehicles manufactured during the late 1970s and early 1980s. It includes 392 vehicles, each described by 9 features covering technical, performance and origin-related characteristics.

The variables are detailed in Table 1.

Variable	Nature	Python type
miles_per_gallon	Quantitative, continuous	float
cylinders	Quantitative, discrete	int
displacement	Quantitative, continuous	float
horsepower	Quantitative, continuous	float
weight_lbs	Quantitative, continuous	float
acceleration	Quantitative, continuous	float
model_year	Quantitative, discrete	int
origin_country	Qualitative, nominal	int
car_name	Qualitative, nominal	object

Table 1: Nature and data type of the automobile attributes

The distribution of vehicles across countries of origin is unbalanced. Indeed, among the 392 vehicles of this dataset, 245 are from USA, 68 from Europe and 79 from Japan. Given that the American vehicles are overrepresented in this dataset (about 62 percent), a country-specific analysis will be performed in addition to the global analysis to capture the trends specific to each class.

1.2 Values

In this section, the following statistical measures are computed for all numerical variables of the dataset: mean, standard deviation, minimum, maximum. It is consistent because it offers an overview of the data, shows the variability of each variable and the presence of potential outliers.

1.2.1 Global statistics

Statistic	MPG	Cylinders	Displacement	Horsepower	Weight (lbs)	Acceleration	Model year
Mean	23.45	5.47	194.41	104.47	2977.58	15.54	75.98
Std	7.81	1.71	104.64	38.49	849.40	2.76	3.68
Min	9.00	3.00	68.00	46.00	1613.00	8.00	70.00
Max	46.60	8.00	455.00	230.00	5140.00	24.80	82.00

Table 2: Descriptive statistics for the full automobile dataset.

The mpg variable, representing the fuel efficiency of vehicles, has a wide range of values: the maximal mpg value is 46.60, and the minimum is 9 mpg (about 5 times less). Since the standard deviation (7.81) is quite large compared to the mean (23.45), the values are quite spread and not strongly concentrated around the average.

The cylinder variable is a discrete variable with a limited range of values, with a minimal and maximal value of respectively 3 and 8.

Displacement, horsepower and weight show similarities and can be analysed together. They have a wide range of values. The standard deviations of respectively 104.64, 38.49 and 849.40 show strong variability in mechanical characteristics of vehicles.

Acceleration values are less sparse, with a standard deviation of only 2.76 for a mean of 15.54, suggesting most vehicles have relatively similar acceleration times.

The model year data is also less sparse than the mechanical data (with a standard deviation of 3.68). The mean value being close to the middle of the interval suggests vehicles are relatively evenly distributed over time.

1.2.2 Statistics by country of origin

The 3 tables below report descriptive statistics computed separately for each country. They show clear differences between countries for each variable.

Statistic	MPG	Cylinders	Displacement	Horsepower	Weight (lbs)	Acceleration	Model year
Mean	20.03	6.28	247.51	119.05	3372.49	14.99	75.59
Std	6.44	1.66	98.38	39.90	795.35	2.74	3.66
Min	9.00	4.00	85.00	52.00	1800.00	8.00	70.00
Max	39.00	8.00	455.00	230.00	5140.00	22.20	82.00

Table 3: Descriptive statistics for American vehicles.

Statistic	MPG	Cylinders	Displacement	Horsepower	Weight (lbs)	Acceleration	Model year
Mean	27.60	4.16	109.63	80.56	2433.47	16.79	75.68
Std	6.58	0.51	22.69	20.16	491.81	3.09	3.42
Min	16.20	4.00	68.00	46.00	1825.00	12.20	70.00
Max	44.30	6.00	183.00	133.00	3820.00	24.80	82.00

Table 4: Descriptive statistics for European vehicles.

Statistic	MPG	Cylinders	Displacement	Horsepower	Weight (lbs)	Acceleration	Model year
Mean	30.45	4.10	102.71	79.84	2221.23	16.17	77.44
Std	6.09	0.59	23.14	17.82	320.50	1.95	3.65
Min	18.00	3.00	70.00	52.00	1613.00	11.40	70.00
Max	46.60	6.00	168.00	132.00	2930.00	21.00	82.00

Table 5: Descriptive statistics for Japanese vehicles.

Firstly, Japanese vehicles have the highest average MPG (30.45), followed by European vehicles (27.60), and American vehicles, with the lowest average MPG value (20.03). There is a clear gap in fuel efficiency between American vehicles and European/Japanese vehicles.

Concerning mechanical characteristics, there are also big differences across countries. American vehicles have the highest average values for the number of cylinders, displacement, horsepower, and weight. European vehicles present intermediate values, while Japanese vehicles are lighter and less powerful on average.

In addition, the standard deviations of these mechanical variables are higher for American vehicles than for Japanese or European ones, indicating more diverse vehicle sizes and configurations for American vehicles.

Concerning the acceleration variable, there are fewer differences between countries. European and Japanese vehicles have average acceleration times of 16.79 and 16.17 seconds, respectively, compared with 14.99 seconds for American vehicles. Since this variable represents the time required to accelerate from 0 to 60 mph, lower values correspond to faster acceleration.

Finally, Japanese vehicles tend to be more recent on average than European or American vehicles.

The global descriptive statistics are closer to those of American vehicles, which was expected because the dataset is dominated by American vehicles.

2 Dataset analysis

2.1 Correlation analysis

Figure 1 presents the correlation matrix computed on the entire dataset.

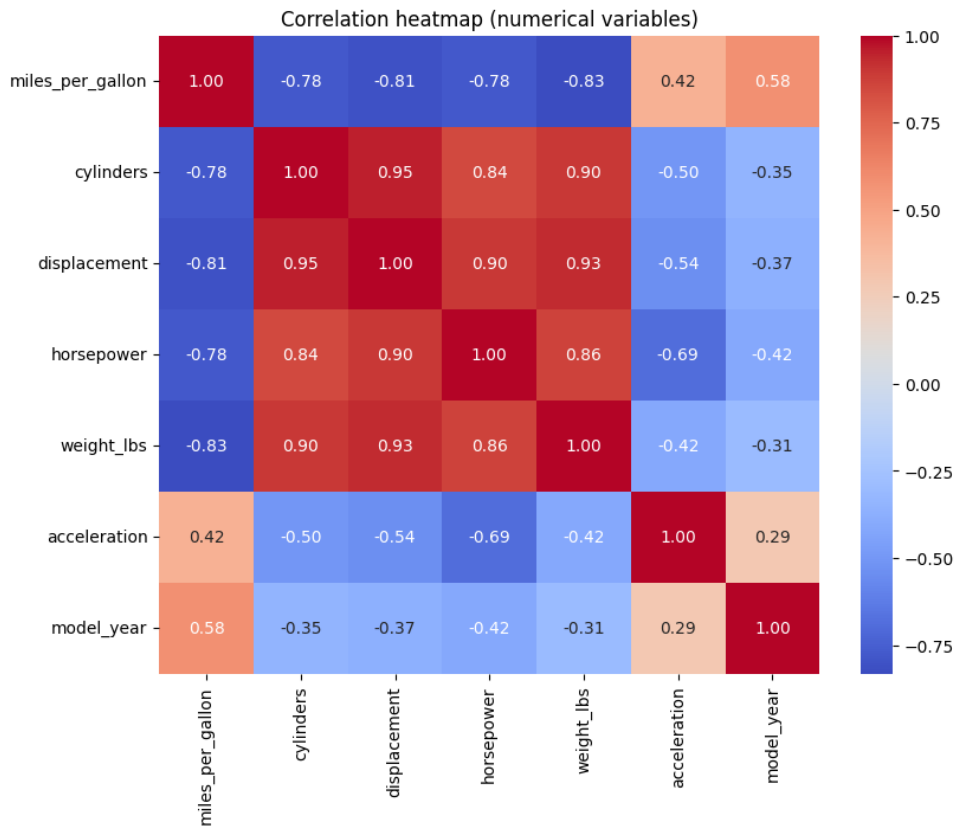


Figure 1: Global correlation matrix of quantitative variables

Strong negative correlations are observed between fuel efficiency and engine-related variables such as weight, displacement and horsepower. Conversely, these engine-related characteristics are strongly positively correlated with each other, indicating a clear structure driven by vehicle size and power.

2.1.1 Correlation analysis by country of origin

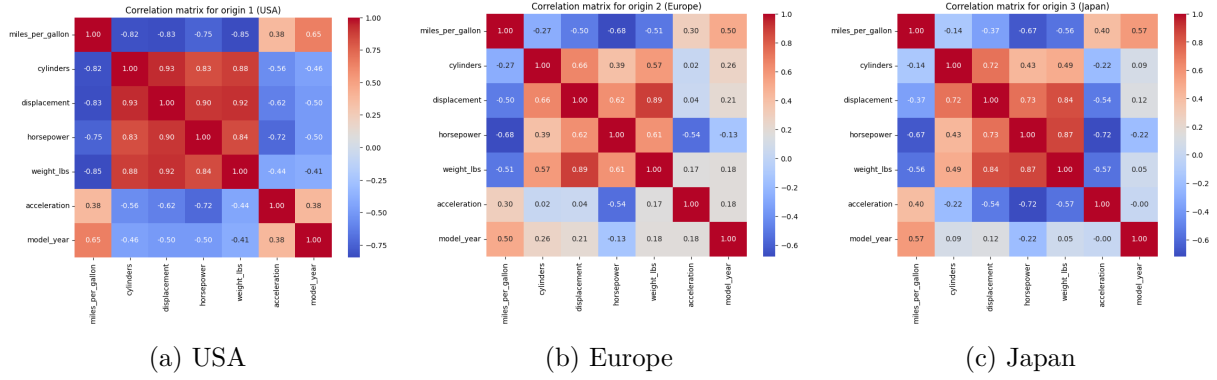


Figure 2: Correlation matrices by country of origin

To investigate whether these relationships differ across countries of origin, correlation matrices were computed separately for each country. American vehicles exhibit stronger correlations between fuel efficiency and mechanical characteristics, reflecting a wider diversity of vehicle sizes. European and Japanese vehicles show weaker correlations, consistent with more homogeneous vehicle designs.

2.2 Pairwise relationships

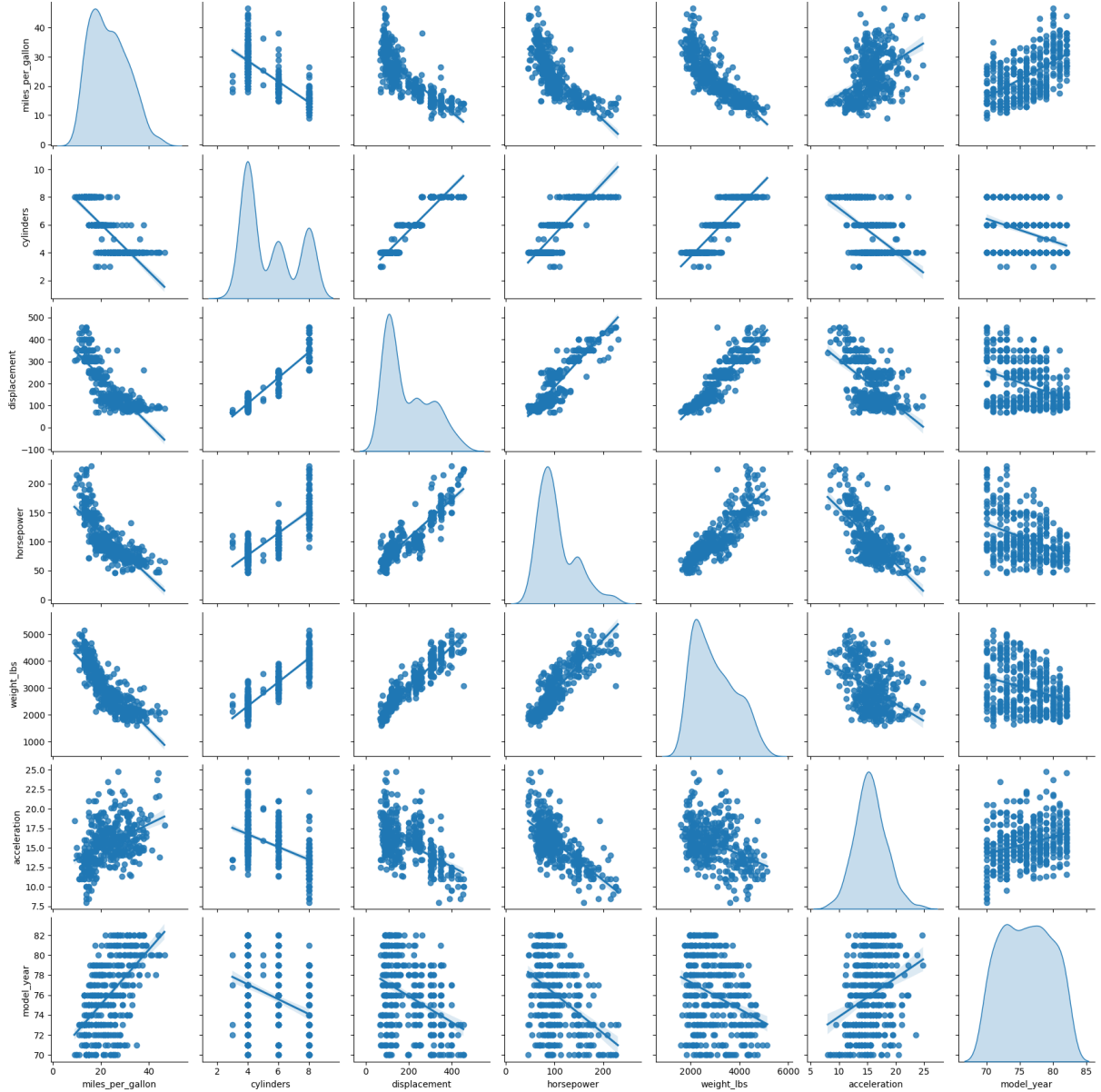


Figure 3: Pairwise relationships of quantitative variables

Figure 3 presents the global pairplot of the quantitative variables. Strong negative relationships are observed between fuel efficiency and mechanical characteristics such as vehicle weight, engine displacement and horsepower. Conversely, these engine-related variables are strongly positively correlated with each other, reflecting a common underlying structure associated with vehicle size and power.

Acceleration exhibits weaker and more dispersed relationships with the other variables, while the model year shows a positive association with fuel efficiency, suggesting the impact of technological progress and regulatory constraints over time.

Although the dataset is dominated by American vehicles, similar trends are observed within each country of origin, indicating that the observed relationships are not solely driven by compositional effects. Pairplots computed separately for each country are provided in Appendix A.

2.3 Distribution Analysis

To assess the distributional properties of selected variables, quantile–quantile (QQ) plots were examined for acceleration and model year. These variables were chosen due to their different natures: acceleration is a continuous performance measure, while model year is a discrete temporal variable.

Figure 4 presents the QQ-plot of the acceleration variable against a normal distribution. The points closely follow the reference line over most of the range, with only moderate deviations in the tails. This suggests that acceleration can be reasonably approximated by a normal distribution, making normality-based interpretations acceptable for this variable.

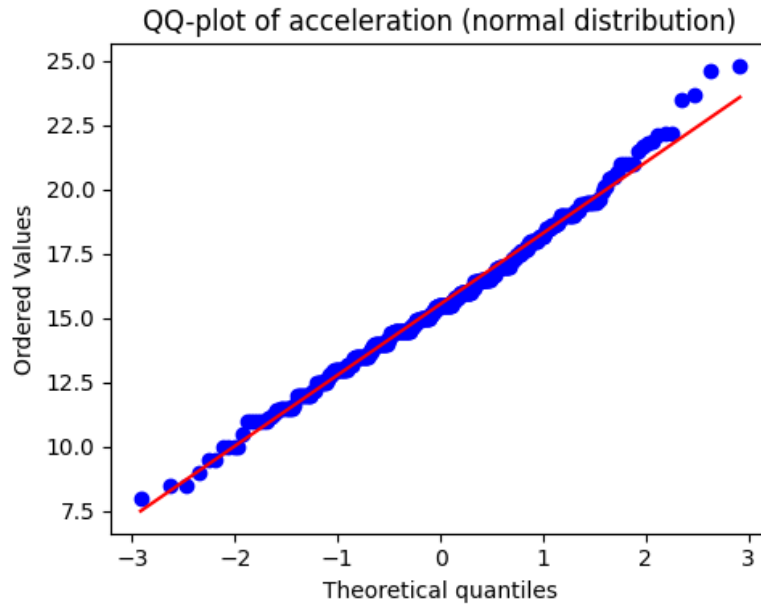


Figure 4: QQ-plot of acceleration against a normal distribution.

Figure 5 shows the QQ-plot of model year against a uniform distribution. The points display an approximately linear pattern with a step-like structure reflecting the discrete nature of the variable. While moderate deviations from perfect uniformity are observed, the distribution of model year appears approximately uniform over the considered time range.

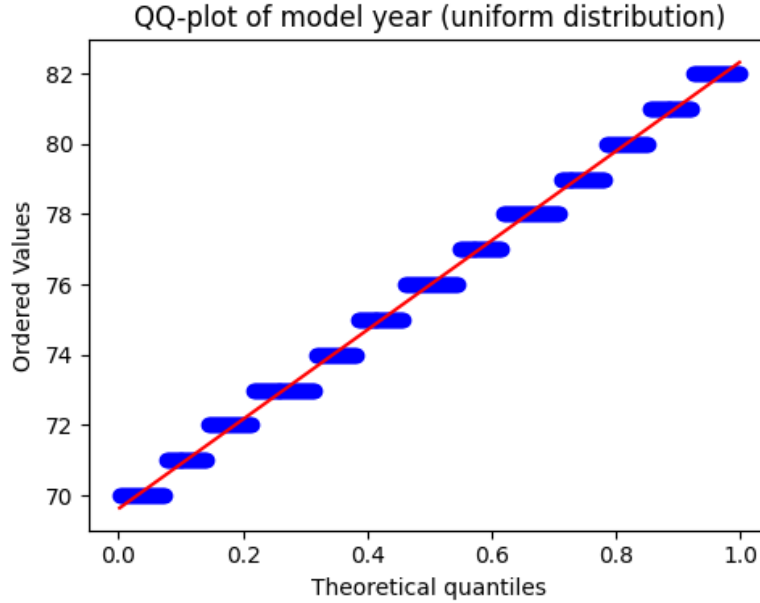


Figure 5: QQ-plot of model year against a uniform distribution.

Overall, these distributional diagnostics highlight that distributional assumptions must be adapted to the nature of each variable. While acceleration exhibits properties compatible with a normal distribution, model year is better described as approximately uniformly distributed over time rather than normally distributed.

2.4 Fuel efficiency standards (CAFE)

In the early 1980s, the US Corporate Average Fuel Economy (CAFE) standard required passenger vehicles to achieve an average fuel efficiency of at least 24 miles per gallon. Based on this regulatory threshold, a binary variable indicating CAFE compliance was introduced.

Figure 6 shows the proportion of vehicles meeting the CAFE standard for each country of origin. Japanese vehicles exhibit the highest compliance rate, followed by European vehicles, while American vehicles display the lowest proportion of compliant cars. This result is consistent with the descriptive statistics, which highlight lower average fuel efficiency for heavier and more powerful American vehicles.

To further investigate the nature of the CAFE threshold, Figure 7 presents the kernel density estimation of fuel efficiency for the entire dataset, together with the 24 mpg threshold. The distribution of miles per gallon is continuous, with no clear separation around the regulatory limit.

A substantial overlap remains between compliant and non-compliant vehicles, indicating that the CAFE standard does not correspond to a natural break in vehicle characteristics but rather imposes an external decision boundary on a continuous spectrum of fuel efficiency.

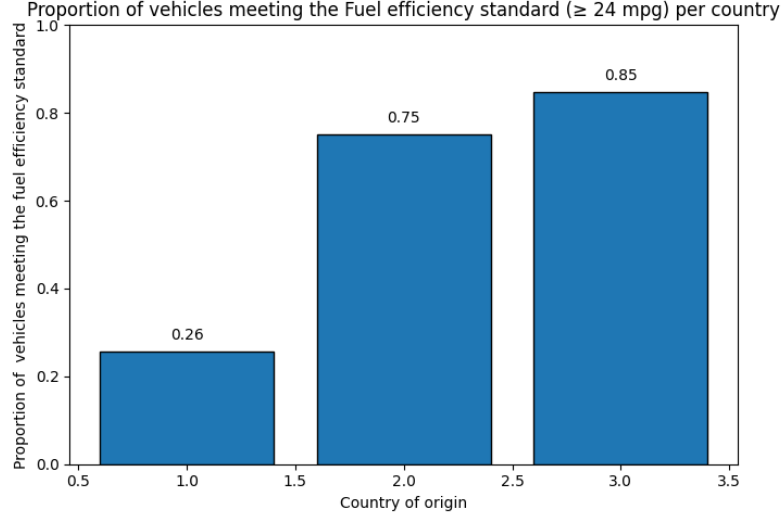


Figure 6: Proportion of vehicles meeting the CAFE standard (≥ 24 mpg) by country of origin.

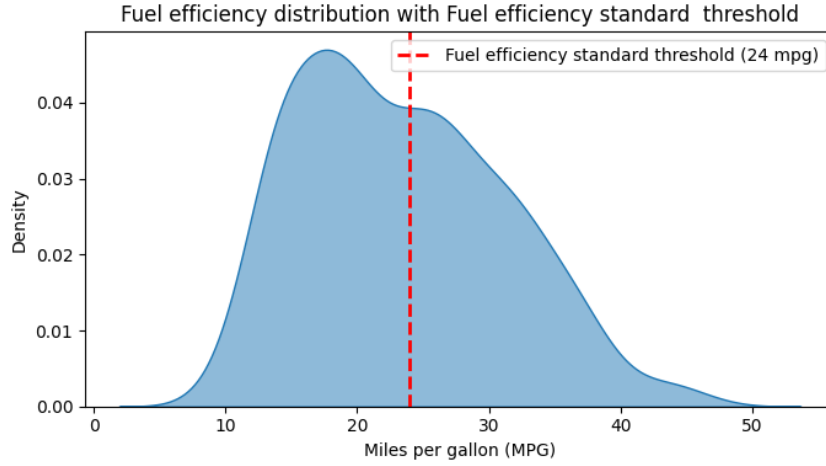


Figure 7: Kernel density estimation of fuel efficiency with the CAFE threshold at 24 mpg.

	MPG	cylinders	displacement	horsepower	weight lbs	acceleration	model year
mean	30.525414	4.132597	112.372928	77.690608	2300.430939	16.486188	77.685083
std	4.986383	0.531338	32.797788	15.425929	374.693018	2.446648	3.672003
min	24.000000	4.000000	68.000000	46.000000	1613.000000	11.300000	70.000000
max	46.600000	8.000000	350.000000	132.000000	3725.000000	24.800000	82.000000

Table 6: Descriptive statistics for fuel-efficient vehicles ($\text{MPG} \geq 24$).

Table 6 summarizes descriptive statistics for fuel-efficient vehicles ($\text{MPG} \geq 24$). These vehicles exhibit a clear mechanical profile characterized by low weight, small engine displacement and reduced horsepower, with an overwhelming predominance of four-cylinder configurations.

Although fuel-efficient vehicles tend to be more recent on average, the wide ranges observed for several variables indicate substantial heterogeneity within this subset. This supports the idea that the fuel efficiency threshold does not correspond to a strict natural separation, but rather defines a compliance group within a continuous spectrum of vehicle designs.

2.5 Principal Component Analysis

To analyze the structure of the automobile dataset and investigate how fuel efficiency relates to other vehicle characteristics, a Principal Component Analysis (PCA) was performed. All variables were standardized prior to the analysis to ensure comparable scales and to prevent variables with larger numerical ranges from dominating the results.

2.5.1 Global PCA including fuel efficiency

A first PCA was conducted using a global set of variables, including mechanical characteristics (number of cylinders, displacement, horsepower and weight), performance (acceleration), production year, and fuel efficiency (miles per gallon). In this analysis, fuel efficiency is treated as a regular variable in order to assess how it integrates within the overall feature space rather than as a response variable.

Figure 8 presents the cumulative explained variance associated with this PCA. The first principal component accounts for a large fraction of the total variance, while the contribution of subsequent components decreases rapidly. This indicates that the dataset is largely structured along a dominant axis, justifying the focus on the first two principal components in the following visualizations.

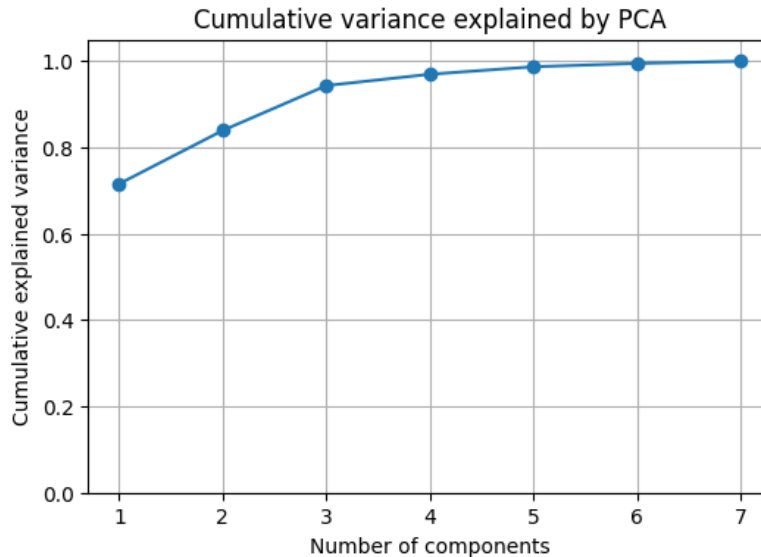


Figure 8: Cumulative explained variance of the global PCA.

To further characterize the distribution of vehicles along the principal axes, Figure 9 shows the pairwise relationships between the first two principal components. The marginal distribution of PC1 exhibits multiple density peaks, highlighting the heterogeneous composition of the dataset. However, the joint distribution of PC1 and PC2 reveals a continuous spread of vehicles rather than a strict separation into distinct groups.

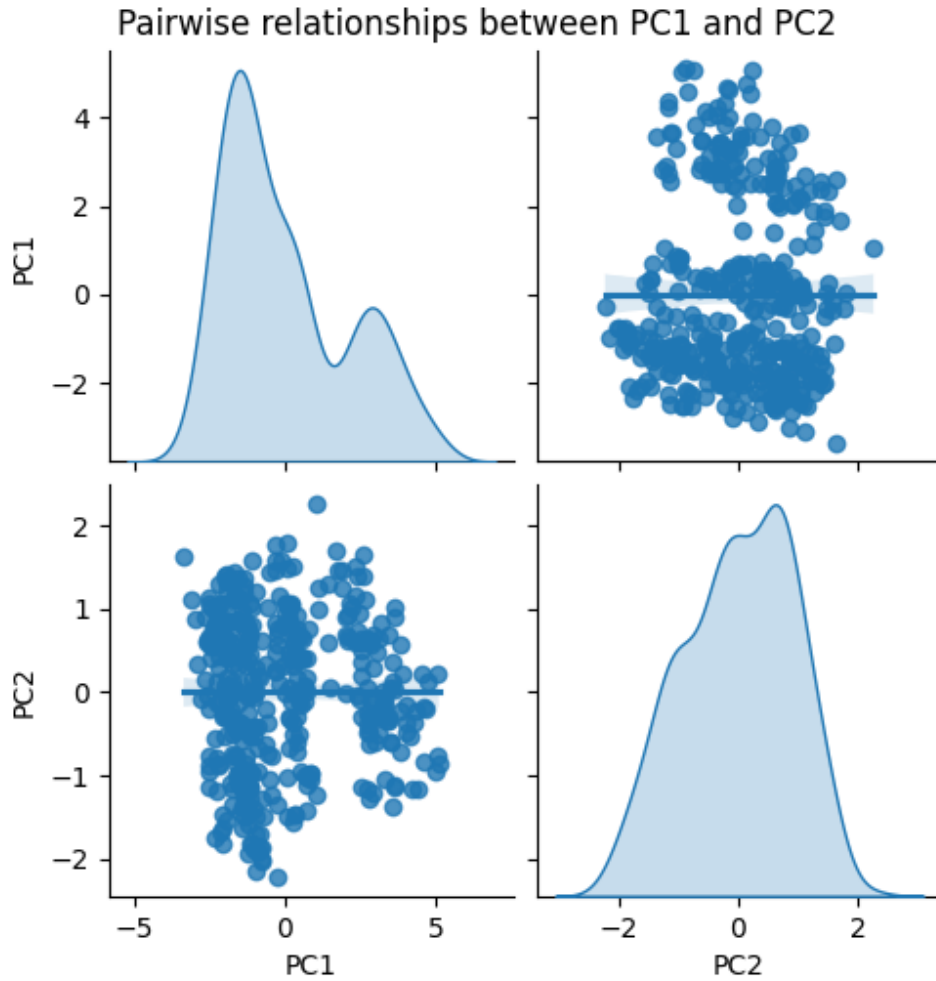


Figure 9: Pairwise relationships and marginal distributions of the first two principal components.

Figure 10 presents the projection of vehicles onto the first two principal components, with points colored according to the country of origin. The first principal component organizes vehicles along a dominant mechanical axis opposing lighter, less powerful vehicles to heavier and more powerful ones. American vehicles are predominantly projected on one side of this axis, while European and Japanese vehicles tend to occupy the opposite side, with substantial overlap between these two regions.

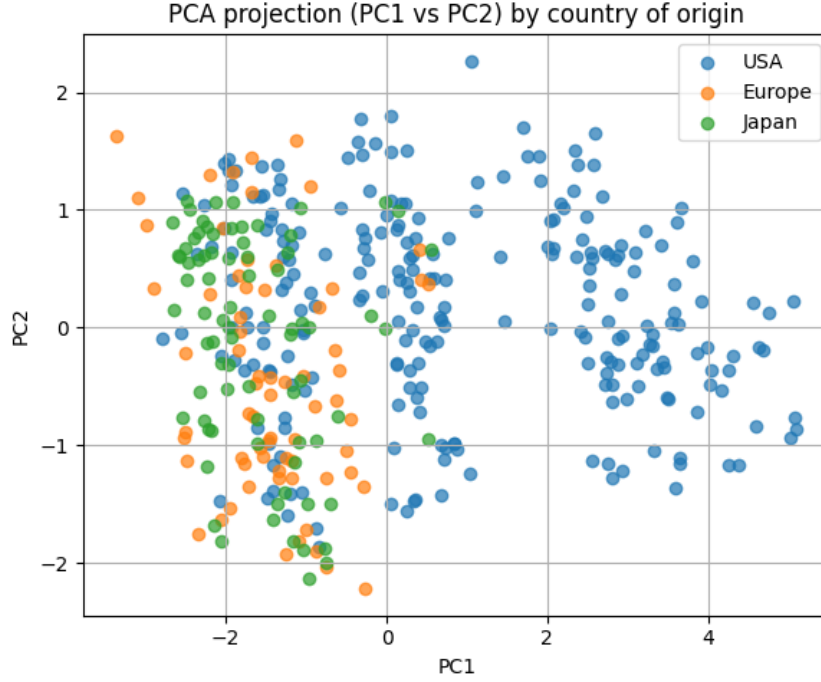


Figure 10: Global PCA projection onto the first two principal components (PC1 vs PC2). Points are colored according to the country of origin.

2.5.2 PCA on mechanical variables

To isolate the intrinsic mechanical structure underlying the observations obtained from the global PCA, a second analysis was performed using strictly mechanical variables only, namely the number of cylinders, engine displacement, horsepower and vehicle weight. Contextual variables such as production year, derived performance indicators such as acceleration, and fuel efficiency itself were deliberately excluded.

The resulting projection onto the first two principal components is shown in Figure 11, with points colored according to the country of origin. The first principal component alone explains more than 90% of the total variance, indicating that vehicle mechanical design is largely governed by a single dominant axis. American vehicles exhibit a wider dispersion along this axis, while European and Japanese vehicles form more compact and overlapping clusters on the opposite side. The analysis of the component loadings provides a clearer interpretation of the two principal axes. For PC1, engine displacement contributes 26.22%, followed by vehicle weight (25.03%), number of cylinders (24.99%) and horsepower (23.76%). The four variables therefore contribute almost equally and all have positive loadings, ranging from 0.487 to 0.512. PC1 can thus be interpreted as an overall mechanical size and power axis, opposing smaller, lighter and less powerful vehicles to larger, heavier and more powerful ones.

PC2 captures a more specific mechanical contrast. Horsepower accounts for 70.64% of this component, with a positive loading of 0.840, while the number of cylinders contributes 24.07% with a negative loading of -0.491. Weight and displacement each contribute less than 3%. PC2 can therefore be interpreted mainly as a contrast between horsepower and engine configuration, distinguishing vehicles with relatively higher horsepower from those characterized by a relatively larger number of cylinders.

This interpretation also provides additional insight into the differences between countries of origin. American vehicles tend to occupy the higher side of PC1, corresponding to larger, heavier and more powerful mechanical configurations. In contrast, European and Japanese vehicles are more concentrated toward the lower side of PC1 and substantially overlap, reflecting generally

smaller, lighter and less powerful designs. American vehicles also exhibit a greater dispersion along PC1, which is consistent with the larger standard deviations previously observed for their mechanical characteristics, particularly weight, displacement and horsepower. This confirms a greater diversity of vehicle sizes and engine configurations within the American subset. Overall, the PCA therefore supports and summarizes the differences between countries already identified through the descriptive statistics.

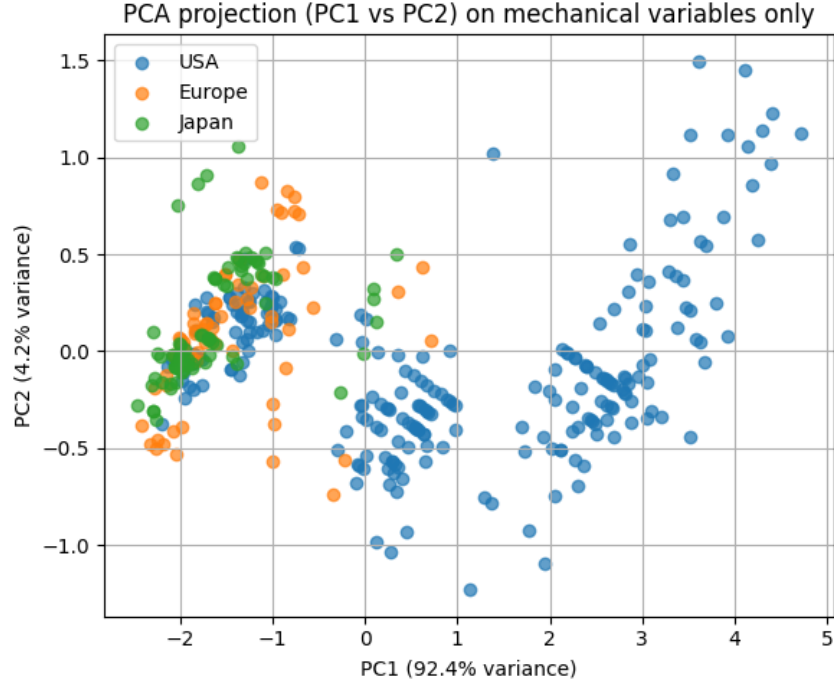


Figure 11: PCA projection (PC1 vs PC2) obtained using mechanical variables only. Points are colored according to the country of origin.

Finally, Figure 12 illustrates the same mechanical PCA projection, with points colored according to fuel efficiency compliance ($\text{MPG} \geq 24$). Fuel-efficient vehicles are predominantly located on the negative side of the first principal component, confirming that fuel efficiency is strongly associated with lighter and less powerful vehicle designs.

Nevertheless, a substantial overlap remains between fuel-efficient and non-fuel-efficient vehicles, indicating that the regulatory threshold does not correspond to a strict mechanical separation but rather defines a compliance category within a continuous design space.

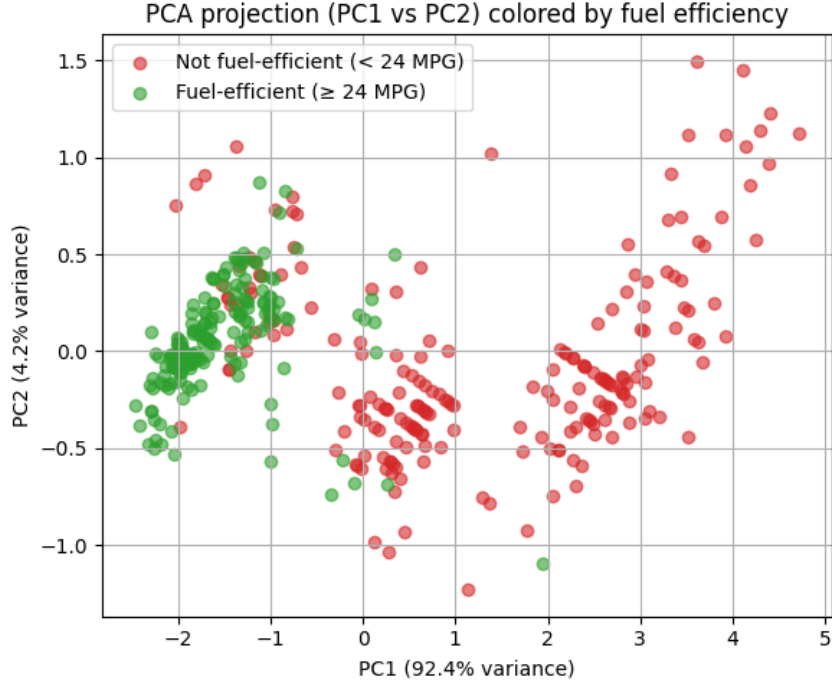


Figure 12: Mechanical PCA projection (PC1 vs PC2) colored according to fuel efficiency compliance.

2.6 Linear Regression Analysis

The PCA highlighted a strong association between fuel efficiency and the dominant mechanical axis of the dataset, which jointly reflects vehicle weight, engine displacement, number of cylinders and horsepower. To quantify this relationship and evaluate its predictive value, two linear regression models were compared.

The dataset was divided into training and test sets, with 80% of the observations used for training and 20% for testing. This allows the models to be evaluated on vehicles that were not used during training.

2.6.1 Regression using vehicle weight

A first linear regression model was trained using vehicle weight as the only explanatory variable:

$$\text{MPG} = \beta_0 + \beta_1 \times \text{Weight} + \varepsilon$$

where β_1 represents the effect of vehicle weight on fuel efficiency.

Figure 13 illustrates the relationship between vehicle weight and fuel efficiency. A clear negative trend is observed: heavier vehicles tend to have lower MPG values.

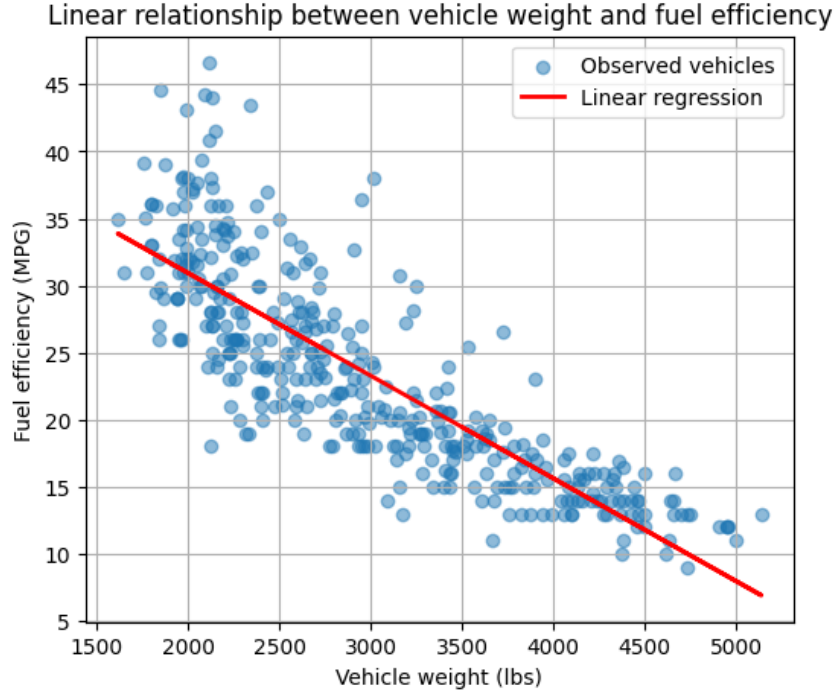


Figure 13: Linear regression between fuel efficiency (MPG) and vehicle weight.

On the test set, the weight-only model achieves a coefficient of determination $R^2 = 0.653$ and a root mean squared error (RMSE) of approximately 4.21 MPG. This confirms that vehicle weight alone contains substantial information about fuel efficiency, although a significant part of the variability remains unexplained.

2.6.2 Multiple linear regression

A second model was then trained using several vehicle characteristics simultaneously: number of cylinders, displacement, horsepower, weight, acceleration and model year.

The multiple linear regression model improves the predictive performance, reaching a test $R^2 = 0.794$ and an RMSE of approximately 3.24 MPG.

Table 7 summarizes the performance of both models.

Model	Test R^2	Test RMSE (MPG)
Weight-only linear regression	0.653	4.21
Multiple linear regression	0.794	3.24

Table 7: Predictive performance of the two linear regression models on the test set.

The improvement obtained with the multiple linear regression shows that vehicle weight, although an important predictor, does not fully explain fuel efficiency. Other mechanical characteristics and model year provide additional predictive information.

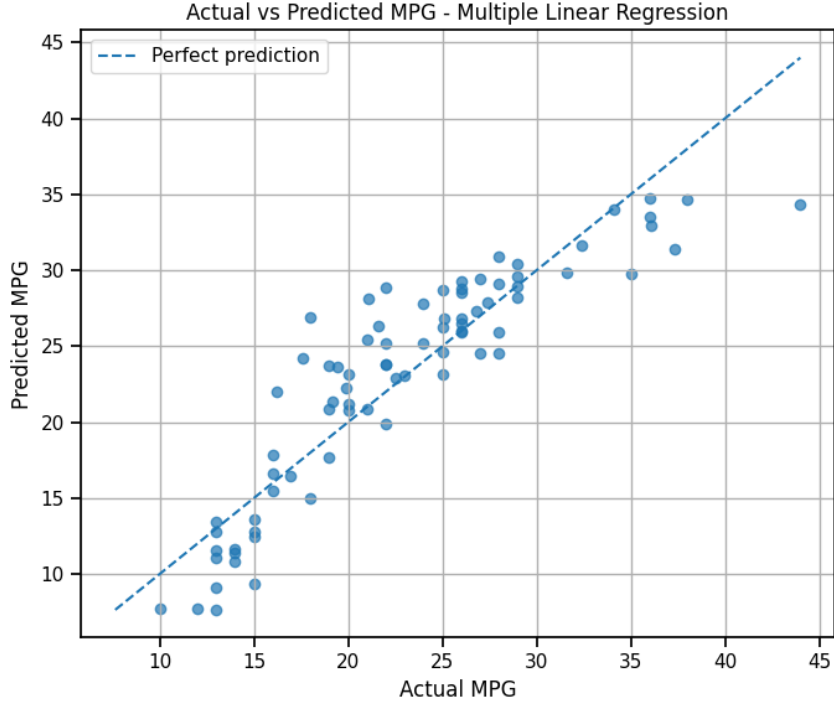


Figure 14: Actual versus predicted MPG for the multiple linear regression model. The dashed line represents perfect predictions.

Figure 14 compares the predicted and observed MPG values on the test set. Most observations remain relatively close to the diagonal, indicating that the model captures a substantial part of the variability in fuel efficiency. Some prediction errors remain, showing that the considered vehicle characteristics do not fully explain fuel consumption.

Overall, these results are consistent with the previous correlation and PCA analyses. Fuel efficiency is strongly associated with the general mechanical characteristics of a vehicle, but its variation cannot be reduced to vehicle weight alone.

3 Conclusion

This study explored the relationships between vehicle characteristics and fuel efficiency in an automobile dataset from the late 1970s and early 1980s.

Fuel efficiency was found to be strongly related to engine size, vehicle weight and power, with notable differences across countries of origin. The PCA confirmed that these relationships are largely structured along a dominant axis associated with vehicle size and mechanical characteristics.

The analysis of CAFE compliance further highlighted systematic trends without revealing a strict separation between vehicle categories. Finally, the regression analysis showed that vehicle weight alone is a strong predictor of MPG, while combining several vehicle characteristics substantially improves predictive performance.

Overall, the results emphasize the multifactorial nature of fuel efficiency and the importance of considering both mechanical and contextual factors.

Appendix A: Pairwise relationships by country of origin

This appendix reports the pairwise relationships between quantitative variables computed separately for each country of origin (1 = USA, 2 = Europe, 3 = Japan). These figures complement

the global pairplot presented in the main text and help assess regional differences in the relationships between fuel efficiency and mechanical characteristics.

A.1 USA

American vehicles show stronger and more dispersed relationships between fuel efficiency and mechanical variables. In particular, fuel efficiency (miles per gallon) decreases markedly with increasing weight, engine displacement and horsepower. This stronger structure is consistent with the higher heterogeneity of American vehicle sizes and engine configurations in the dataset.

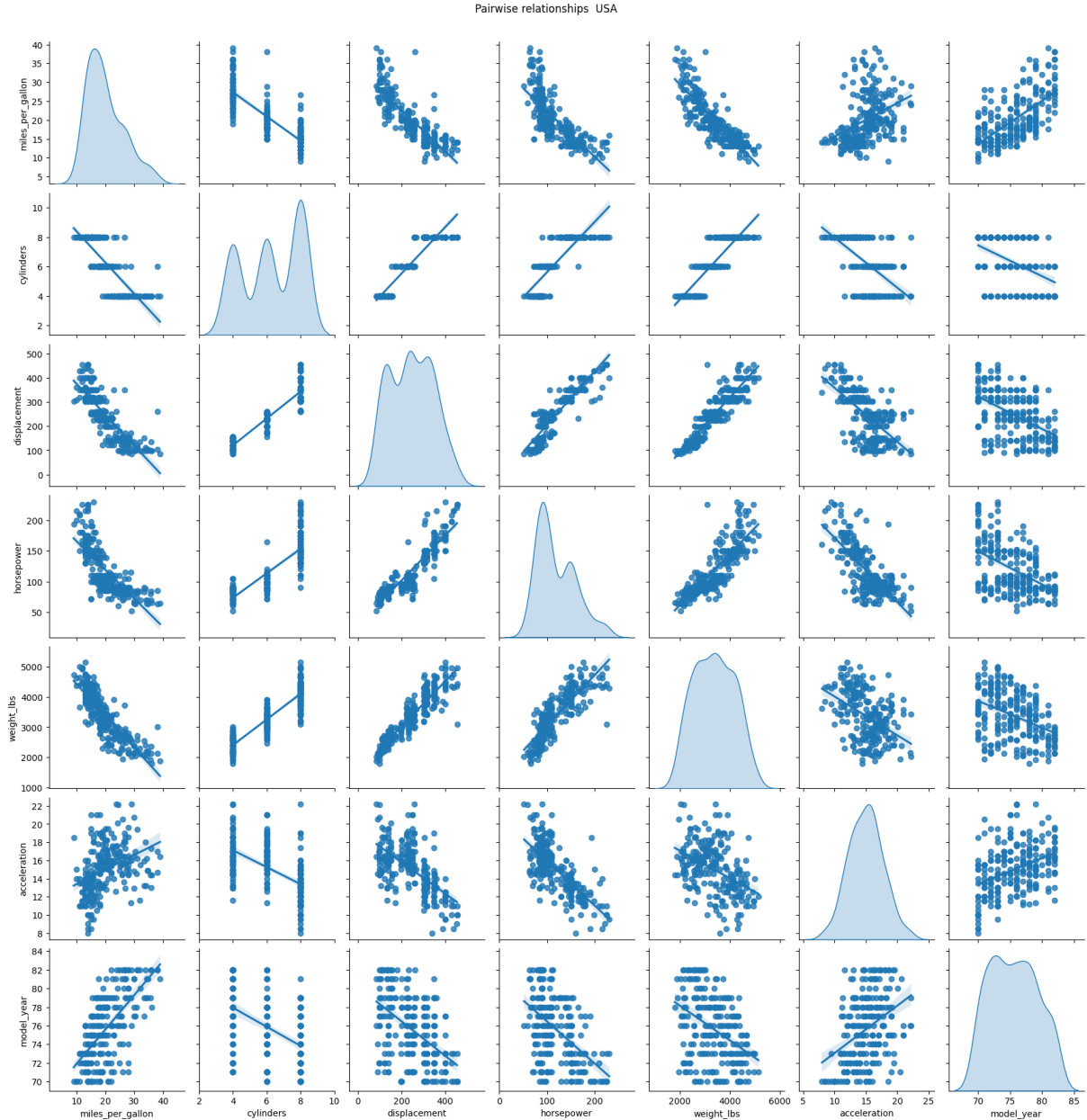


Figure 15: Pairplot of quantitative variables for American vehicles (origin_country = 1).

A.2 Europe

European vehicles exhibit weaker pairwise relationships compared to the global dataset. While fuel efficiency still tends to decrease with weight and engine power, the overall variability is lower,

reflecting a more homogeneous range of vehicle designs. As a consequence, discrete variables such as the number of cylinders provide less discrimination within this subset.

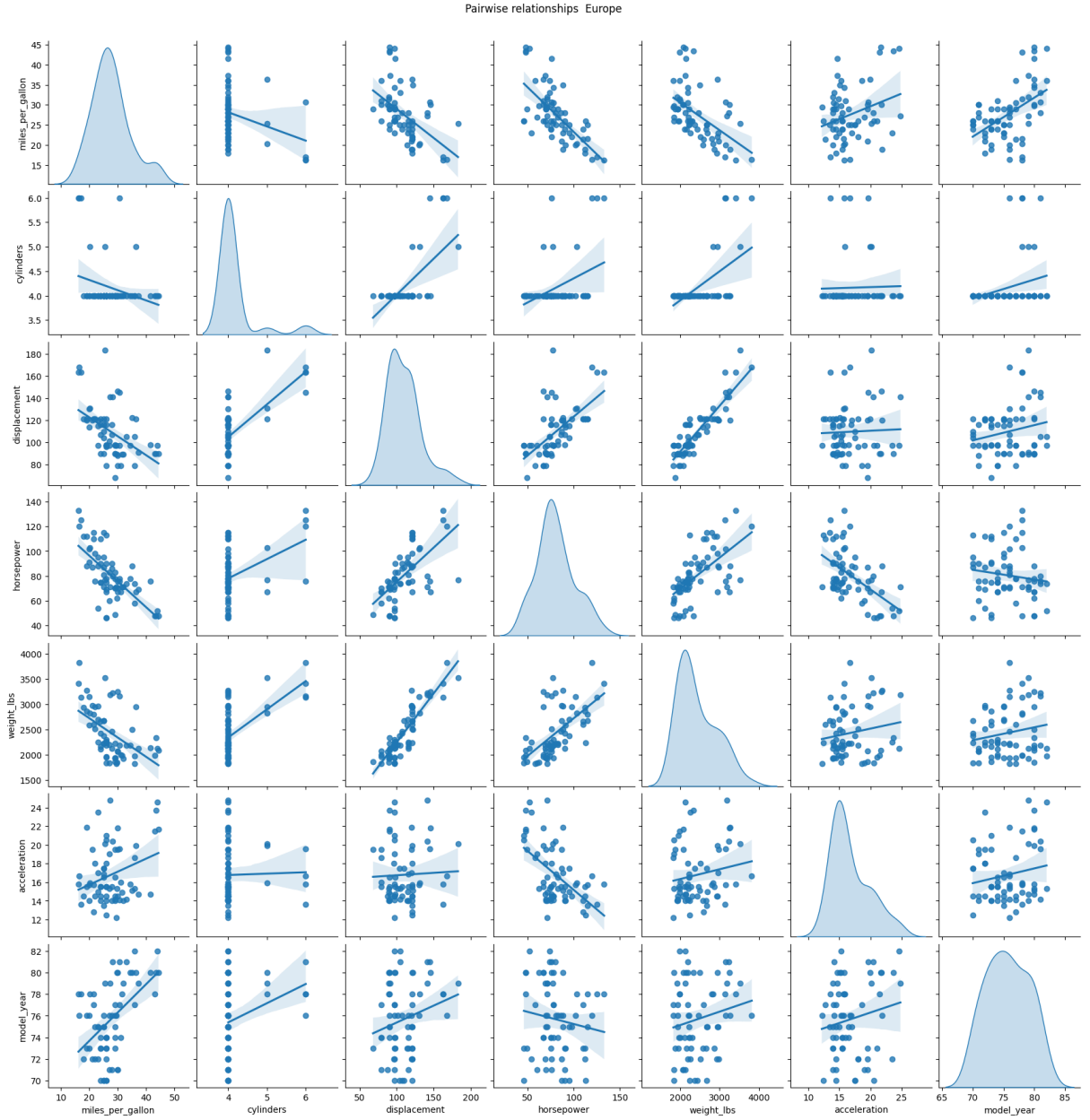


Figure 16: Pairplot of quantitative variables for European vehicles (`origin_country = 2`).

A.3 Japan

Japanese vehicles tend to combine high fuel efficiency with relatively light weight and smaller engine sizes. Pairwise relationships remain visible, but the reduced variability in several mechanical variables yields more compact clouds of points. Overall, the Japanese subset appears strongly oriented toward efficiency, consistent with the descriptive statistics and country comparisons reported in the main text.

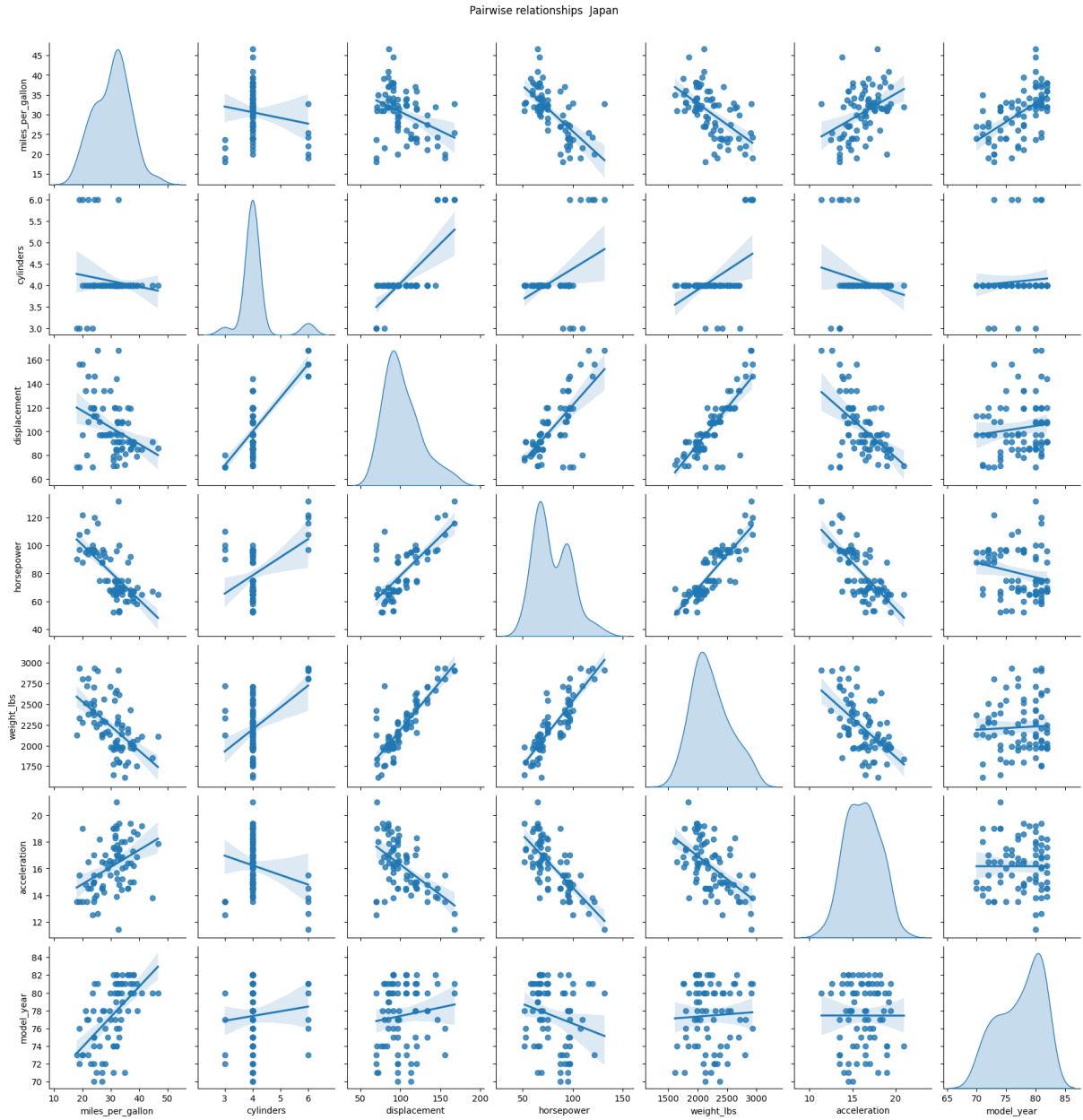


Figure 17: Pairplot of quantitative variables for Japanese vehicles (`origin_country = 3`).